

9.1 Why fluid models?

An online seller makes decisions before future demand is known. A clairvoyant who sees the entire future can do better, but solving a separate hindsight problem for every possible future is not a practical control method. This raises our first question:

How can a deterministic optimization problem provide a useful benchmark for a stochastic online system?

The answer begins by relaxing the hindsight allocation problem and then asking how its value changes with demand. Concavity and Jensen's inequality let us replace random demand by its mean while preserving an upper bound.

Our second question is operational:

How does the benchmark extend when a request consumes several shared resources simultaneously?

That extension produces a network fluid LP. Its dual variables price scarce resources, while its primal solution predicts the aggregate actions of hindsight. A static randomized policy then tries to realize those primal counts. The progression is

hindsight allocation $\longrightarrow$ LP relaxation $\longrightarrow$ mean demand and Jensen
 $\longrightarrow$ network fluid LP $\longrightarrow$ bid prices and predicted action counts
 $\longrightarrow$ static randomized implementation $\longrightarrow O(\sqrt{T})$ loss.

9.2 Hindsight allocation with one resource

There is one resource with capacity B and a finite set of request types $\sigma \in \mathcal{S}$. Accepting a type- σ request earns reward v_σ and consumes one unit. Let

N_σ = the random number of type- σ arrivals over the horizon

and write $N = (N_\sigma)_{\sigma \in \mathcal{S}}$. Capital letters denote the random counts; a realized count vector is denoted by n .

After seeing a realization n , an offline clairvoyant solves

$$\begin{aligned} \text{OPT}(n) = \max_{x \in \mathbb{Z}_+^{|\mathcal{S}|}} \quad & \sum_{\sigma \in \mathcal{S}} v_\sigma x_\sigma \\ \text{s.t.} \quad & \sum_{\sigma \in \mathcal{S}} x_\sigma \leq B, \\ & 0 \leq x_\sigma \leq n_\sigma \quad \text{for every } \sigma \in \mathcal{S}. \end{aligned} \tag{9.1}$$

The decision x_σ is the number of type- σ requests accepted by hindsight.

Relax integrality while retaining the realized demand bounds:

$$\begin{aligned} \text{LP}_H(n) = \max_{x \geq 0} \quad & \sum_{\sigma \in \mathcal{S}} v_\sigma x_\sigma \\ \text{s.t.} \quad & \sum_{\sigma \in \mathcal{S}} x_\sigma \leq B, \\ & x_\sigma \leq n_\sigma \quad \text{for every } \sigma \in \mathcal{S}. \end{aligned} \tag{9.2}$$

This LP is easier to analyze, but it is still clairvoyant: its right-hand side contains the future realization n . Relaxation gives the pathwise relation

$$\text{OPT}(n) \leq \text{LP}_H(n), \quad \text{and hence} \quad \mathbb{E}[\text{OPT}(N)] \leq \mathbb{E}[\text{LP}_H(N)]. \tag{9.3}$$

9.3 Jensen and concavity of the LP value

For a concave function f and a finite-support random variable X , Jensen's inequality states

$$\mathbb{E}[f(X)] \leq f(\mathbb{E}[X]). \tag{9.4}$$

Geometrically, a chord lies below a concave graph. Averaging points on the graph therefore lies below evaluating the graph at the average input. For a convex function, the inequality reverses.

The board examples give three quick checks on the direction:

$$\begin{aligned} f(x) = x^2 &\implies \mathbb{E}[X^2] \geq \mathbb{E}[X]^2, && \text{because } f \text{ is convex,} \\ g(x) = \sqrt{x} &\implies \mathbb{E}[\sqrt{X}] \leq \sqrt{\mathbb{E}[X]}, && \text{because } g \text{ is concave,} \\ g_c(x) = \min\{x, c\} &\implies \mathbb{E}[\min\{X, c\}] \leq \min\{\mathbb{E}[X], c\}, && \text{because } g_c \text{ is concave.} \end{aligned}$$

Here $X \geq 0$ in the square-root example. By contrast, $h_c(x) = \max\{x, c\}$ is convex and its Jensen inequality points in the opposite direction. These truncated linear functions are useful

mental models for capacity: additional demand helps until a scarce resource is full, after which its marginal value can fall to zero.

The needed question is now precise:

Is $\text{LP}_H(n)$ concave as a function of the demand upper bound n ?

9.3.1 Proof by mixing feasible allocations

Take demand vectors n^1, n^2 and optimal LP solutions x^1, x^2 . For $\theta \in [0, 1]$, define

$$n^\theta = \theta n^1 + (1 - \theta)n^2, \quad x^\theta = \theta x^1 + (1 - \theta)x^2.$$

Capacity feasibility is preserved:

$$\sum_{\sigma} x_{\sigma}^{\theta} = \theta \sum_{\sigma} x_{\sigma}^1 + (1 - \theta) \sum_{\sigma} x_{\sigma}^2 \leq B.$$

Demand feasibility is also preserved:

$$0 \leq x_{\sigma}^{\theta} \leq \theta n_{\sigma}^1 + (1 - \theta)n_{\sigma}^2 = n_{\sigma}^{\theta}.$$

Thus x^{θ} is feasible for demand bound n^{θ} . The optimum at n^{θ} is at least the value of this feasible solution, so

$$\begin{aligned} \text{LP}_H(n^{\theta}) &\geq \sum_{\sigma} v_{\sigma} x_{\sigma}^{\theta} \\ &= \theta \text{LP}_H(n^1) + (1 - \theta) \text{LP}_H(n^2). \end{aligned}$$

This is exactly concavity.

9.3.2 Proof from LP duality

Associate $\rho \geq 0$ with the capacity constraint and $\mu_{\sigma} \geq 0$ with the demand bound $x_{\sigma} \leq n_{\sigma}$. The dual of (9.2) is

$$\begin{aligned} \text{LP}_H(n) &= \min_{\rho, \mu \geq 0} \quad \rho B + \sum_{\sigma \in \mathcal{S}} \mu_{\sigma} n_{\sigma} \\ \text{s.t.} \quad &\rho + \mu_{\sigma} \geq v_{\sigma} \quad \text{for every } \sigma \in \mathcal{S}. \end{aligned} \tag{9.5}$$

For each fixed feasible (ρ, μ) , the objective is affine in n . The LP value is the pointwise minimum over these affine functions. A pointwise minimum of affine functions is concave.

The two proofs emphasize different ideas. Feasible-solution mixing is direct and extends immediately to networks. The dual proof shows why shadow prices create the supporting affine functions in Figure 9.1.

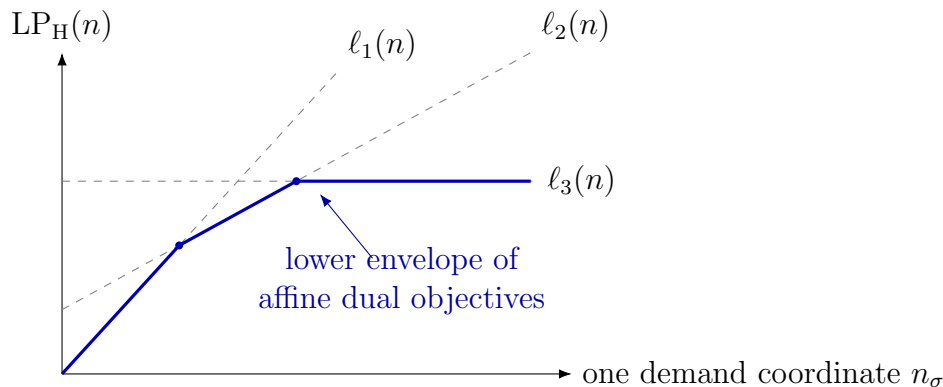


Figure 9.1. Holding the other demand coordinates fixed, the LP value is the lower envelope of feasible dual objectives. Its slopes decrease as demand grows, which is the geometry of a concave piecewise-linear function.

9.4 The fluid upper bound

Define the mean demand vector

$$d = \mathbb{E}[N]$$

and the *fluid LP*

$$\text{FLUID} := \text{LP}_H(d) = \text{LP}_H(\mathbb{E}[N]). \quad (9.6)$$

This is also called a deterministic or certainty-equivalent LP: it replaces random demand by its expectation.

Concavity and Jensen give

$$\mathbb{E}[\text{LP}_H(N)] \leq \text{LP}_H(\mathbb{E}[N]) = \text{FLUID}. \quad (9.7)$$

Combining information, relaxation, and Jensen yields the central benchmark chain. For any feasible online policy π with random revenue R^π ,

$$\boxed{\mathbb{E}[R^\pi] \leq \mathbb{E}[\text{OPT}(N)] \leq \mathbb{E}[\text{LP}_H(N)] \leq \text{LP}_H(\mathbb{E}[N]) = \text{FLUID}.} \quad (9.8)$$

Each step has a different reason:

- $\mathbb{E}[R^\pi] \leq \mathbb{E}[\text{OPT}(N)]$ comes from the clairvoyant's extra information;
- $\text{OPT}(N) \leq \text{LP}_H(N)$ comes from relaxing integrality;
- $\mathbb{E}[\text{LP}_H(N)] \leq \text{LP}_H(\mathbb{E}[N])$ comes from concavity and Jensen.

Remark 9.1 (The direction is not automatic). *It is not generally true that optimizing at expected data upper-bounds the expected value of optimizing at random data. The direction in (9.7) depends on concavity of this LP value as a function of the demand right-hand side.*

9.5 The classroom example: realized demand versus fluid demand

Consider the board example with horizon $T = 8$ and capacity $B = 4$. Let V_t be the random value of arrival t . The realized request values are

$$(v_1, \dots, v_8) = (3, 3, 2, 1, 2, 1, 1, 3).$$

The offline clairvoyant sorts the values and accepts three value-3 requests and one of the two value-2 requests. If it breaks the value-2 tie in favor of the earlier arrival, one corresponding action vector is

$$a^H = (1, 1, 1, 0, 0, 0, 0, 1), \quad \text{OPT} = 3 + 3 + 2 + 3 = 11.$$

Because there are only three request types, the same hindsight problem can be written in terms of aggregate counts. Let N_σ be the number of value- σ arrivals, for $\sigma \in \{1, 2, 3\}$. On this sample path,

$$n = (n_1, n_2, n_3) = (3, 2, 3).$$

The realized hindsight LP is

$$\begin{aligned} \max \quad & x_1 + 2x_2 + 3x_3 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 \leq 4, \\ & 0 \leq x_\sigma \leq n_\sigma \end{aligned} \quad (\sigma = 1, 2, 3).$$

It chooses $x^H = (0, 1, 3)$ and has value 11. In this single-resource problem the LP relaxation is integral, so the hindsight LP and hindsight integer program agree.

Now let $q_\sigma = \mathbb{P}(V_t = \sigma)$ be the model probability of type σ . Then $\mathbb{E}[N_\sigma] = Tq_\sigma$, and the fluid LP can be written either in counts or in per-period rates:

$$\begin{aligned} \text{FLUID} &= \text{LP}_H(Tq) \\ &= T \max_{y \geq 0} \left\{ \sum_{\sigma=1}^3 \sigma y_\sigma : \sum_{\sigma=1}^3 y_\sigma \leq \frac{B}{T}, \quad 0 \leq y_\sigma \leq q_\sigma \right\}. \end{aligned}$$

For example, if $Tq_3 \leq B$ and expected type-2 demand is large enough to fill the remainder, the fluid solution has

$$x_3^{\text{fl}} = Tq_3, \quad x_2^{\text{fl}} = B - Tq_3, \quad x_1^{\text{fl}} = 0.$$

The two LP objects are therefore different:

$\text{LP}_H(N)$: clairvoyant LP evaluated at realized random demand, $\text{LP}_H(\mathbb{E}[N])$: deterministic fluid LP evaluated at mean demand.

The fluid value need not dominate the realized hindsight value on every sample path. Its guarantee in (9.8) is an expectation bound.

9.6 From one resource to a network

Many products use several resources at once:

- a connecting airline itinerary uses a seat on each flight leg;
- a hotel stay uses one room on several nights;
- an order can consume several inventory components;
- an ad allocation can draw from several campaign budgets.

One scalar protection level no longer describes scarcity. We need a price for each resource and a way to combine those prices for a request that uses several of them.

There are m resources with remaining capacity

$$b = (b_1, \dots, b_m) \in \mathbb{Z}_+^m$$

and k request types. A type- j request earns reward r_j and consumes resource vector

$$a_j \in \mathbb{Z}_+^m.$$

Collect the columns into $A = [a_1 \ \dots \ a_k]$. With τ periods remaining, let the random future count of type j be

$$Z_j(\tau) = \sum_{s=1}^{\tau} \mathbf{1}\{J_s = j\}, \quad \mathbb{E}[Z_j(\tau)] = \tau \lambda_j.$$

Write $Z = (Z_1, \dots, Z_k)$ and $d = \mathbb{E}[Z]$ when the remaining horizon is clear.

9.7 A running two-leg airline example

There are two flight legs with one seat on each. Products 1 and 2 are local itineraries worth 6; product 3 is a connecting itinerary worth 9.

The consumption vectors are

$$a_1 = (1, 0), a_2 = (0, 1), a_3 = (1, 1).$$

If the expected future count of every product is one, the fluid LP is

$$\begin{aligned} \max \quad & 6x_1 + 6x_2 + 9x_3 \\ \text{s.t.} \quad & x_1 + x_3 \leq 1, \\ & x_2 + x_3 \leq 1, \\ & 0 \leq x_j \leq 1 \quad (j = 1, 2, 3). \end{aligned}$$

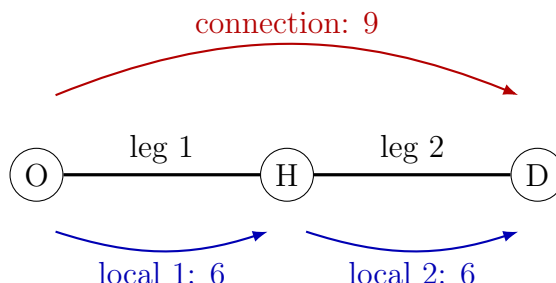


Figure 9.2. A connecting itinerary uses both scarce legs. Its reward must be compared with the combined opportunity cost of both seats, not with either local reward separately.

Its solution is

$$x^* = (1, 1, 0), \quad \text{FLUID} = 12.$$

The connecting itinerary has the largest individual reward, but it consumes two resources whose combined use can generate 12 from the local products.

9.8 Network hindsight and the fluid LP

For realized future counts z , the network clairvoyant solves

$$\begin{aligned} \text{OPT}(z, b) = \max_{x \in \mathbb{Z}_+^k} \quad & r^\top x \\ \text{s.t.} \quad & Ax \leq b, \\ & 0 \leq x \leq z. \end{aligned} \tag{9.9}$$

Relax integrality and replace the random demand bounds by their means to obtain

$$\begin{aligned} \text{FLUID}(d, b) = \max_{x \geq 0} \quad & r^\top x \\ \text{s.t.} \quad & Ax \leq b, \\ & x \leq d. \end{aligned} \tag{9.10}$$

The one-resource concavity proof was not special to one resource. If x^1 and x^2 satisfy $Ax^k \leq b$ and $x^k \leq d^k$, then their convex combination satisfies the same capacity vector b and the mixed demand bound $\theta d^1 + (1 - \theta)d^2$. Thus the relaxed network value is concave in d . After the integrality relaxation, Jensen again gives

$$\mathbb{E}[\text{OPT}(Z, b)] \leq \text{FLUID}(\mathbb{E}[Z], b). \tag{9.11}$$

9.9 Duality and bid prices

Associate $p \geq 0$ with the resource constraints $Ax \leq b$ and $\eta \geq 0$ with the demand bounds $x \leq d$. The dual of (9.10) is

$$\begin{aligned} \min_{p, \eta \geq 0} \quad & b^\top p + d^\top \eta \\ \text{s.t.} \quad & a_j^\top p + \eta_j \geq r_j \quad \text{for } j = 1, \dots, k. \end{aligned} \tag{9.12}$$

The resource dual variable p_i is the fluid shadow price of one additional unit of resource i . Revenue management calls it a *bid price*.

For the two-leg example, one optimal dual solution is

$$p = (5, 5), \quad \eta = (1, 1, 0).$$

It is feasible, and its objective value is

$$b^\top p + d^\top \eta = 5 + 5 + 1 + 1 = 12,$$

equal to the primal value. The resource costs are

product	$p^\top a_j$	r_j
1	5	6
2	5	6
3	10	9.

Each local itinerary pays more than the approximate opportunity cost of its one seat. The connection pays less than the combined opportunity cost of both seats. This motivates the bid-price control

$$\boxed{\text{accept type } j \text{ if it is feasible and } r_j \geq p^\top a_j.} \tag{9.13}$$

The LP has turned one centralized allocation calculation into a decentralized accept/reject rule for arriving requests.

9.10 Exact versus approximate opportunity costs

This course has now encountered prices and opportunity costs in four forms:

Lecture 1.

Dual prices coordinated buyers competing for heterogeneous goods.

Lecture 6.

Littlewood's rule priced the expected displacement caused by using one unit now.

Lectures 7–8.

Fluid quantities scale like T , while stochastic deviations occur on the $\sqrt{T}$ scale.

Lectures 9–10.

Network LP shadow prices approximate the opportunity cost of several shared resources at once.

The exact dynamic opportunity cost of accepting type j with t periods and capacity vector b is

$$v_{t-1}(b) - v_{t-1}(b - a_j).$$

It is state-dependent and exact, but computing it requires a dynamic program on a multidimensional state space. The fluid approximation is

$$p^\top a_j.$$

It is static and approximate, but decomposes across resources and scales to large networks:

$\begin{aligned} \text{exact dynamic marginal value} &= v_{t-1}(b) - v_{t-1}(b - a_j), \\ \text{fluid approximation} &= p^\top a_j. \end{aligned}$
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9.11 Primal counts as predictions

The dual is only half of the fluid LP's information. Its primal solution x^fl predicts how many requests of each type a relaxed offline optimum would accept if future demand equaled its mean. Thus the two sides of the same LP support two distinct uses:

$\begin{aligned} \text{dual variables } p &: \text{ approximate resource opportunity costs,} \\ \text{primal solution } x^\text{fl} &: \text{ predicted aggregate hindsight actions.} \end{aligned}$
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The classical policy uses the first interpretation: solve the LP, extract bid prices, and apply (9.13). We now use the second interpretation to build the simplest possible implementation of the fluid plan.

9.12 A static randomized fluid policy

Return to the time-zero network fluid LP. Suppose there are T iid arrival opportunities, type j arrives with probability q_j , and x_j^fl is an optimal fluid count. For every type with $q_j > 0$, define

$$\alpha_j = \frac{x_j^\text{fl}}{Tq_j}. \tag{9.14}$$

The demand constraint $x_j^{\text{fl}} \leq Tq_j$ guarantees that $\alpha_j \in [0, 1]$. If $q_j = 0$, set $\alpha_j = 0$.

The *static randomized fluid policy* is:

1. solve the fluid LP once at time zero;
2. whenever type j arrives, independently mark it for acceptance with probability α_j ;
3. accept a marked request if its required resources remain available, and otherwise reject it.

The probabilities never change with time or state. The policy is static even though feasibility is checked online.

Let X_j be the number of type- j requests marked for acceptance before the capacity check. In each period, a type- j mark occurs with probability $q_j\alpha_j = x_j^{\text{fl}}/T$. Hence

$$X_j \sim \text{Binomial}\left(T, \frac{x_j^{\text{fl}}}{T}\right), \quad \mathbb{E}[X_j] = x_j^{\text{fl}}, \quad (9.15)$$

and

$$\text{Var}(X_j) = x_j^{\text{fl}} \left(1 - \frac{x_j^{\text{fl}}}{T}\right) \leq x_j^{\text{fl}}. \quad (9.16)$$

Thus the policy implements the fluid counts in expectation. Its realized counts still fluctuate around those targets.

9.13 Why the loss has square-root scale

There are two mismatches for each type:

$$X_j < x_j^{\text{fl}} \quad (\text{undershoot}), \quad X_j > x_j^{\text{fl}} \quad (\text{overshoot}).$$

By Cauchy–Schwarz and (9.16),

$$\mathbb{E} |X_j - x_j^{\text{fl}}| \leq \sqrt{\text{Var}(X_j)} \leq \sqrt{x_j^{\text{fl}}}. \quad (9.17)$$

The next theorem states a transparent network version. It uses the common airline-incidence assumption $a_{ij} \in \{0, 1\}$: a product either uses one unit of a resource or does not use it. Let

$$k = \text{number of request types}, \quad a_{\max} = \max_j \sum_i a_{ij}, \quad r_{\max} = \max_j r_j.$$

Thus $a_{\max}$ is the largest number of resources used by one request.

Theorem 9.2 (Static fluid policy). Suppose $a_{ij} \in \{0, 1\}$, every request uses at least one resource, rewards are nonnegative and at most $r_{\max}$, and arrivals are iid from a fixed set of k types. Let R^{static} be the reward of the policy defined in (9.14). Then

$$\begin{aligned} \text{FLUID} - \mathbb{E}[R^{\text{static}}] &\leq r_{\max}(1 + a_{\max}) \sum_{j=1}^k \mathbb{E} |X_j - x_j^{\text{fl}}| \\ &\leq r_{\max}(1 + a_{\max}) \sum_{j=1}^k \sqrt{x_j^{\text{fl}}} \\ &\leq r_{\max}(1 + a_{\max}) \sqrt{kT}. \end{aligned} \tag{9.18}$$

In particular, for a fixed finite-type network, the additive loss from the fluid upper bound is $O(\sqrt{T})$.

Proof: Undershooting type j can lose at most $r_j(x_j^{\text{fl}} - X_j)^+$. Overshooting may cause marked requests to be blocked by capacity. Charge each blocked request to one resource that was already full. For a binary incidence matrix, the number of such charges to resource i is at most

$$\left(\sum_j a_{ij} X_j - B_i \right)^+ \leq \sum_j a_{ij} (X_j - x_j^{\text{fl}})^+,$$

where the inequality uses feasibility of the fluid solution, $\sum_j a_{ij} x_j^{\text{fl}} \leq B_i$. Each blocked request is worth at most $r_{\max}$. Summing the undershoot and blocking charges gives the first line of (9.18). Equation (9.17) gives the second. Finally, $\sum_j x_j^{\text{fl}} \leq T$ because at most one request arrives per period, so Cauchy–Schwarz gives $\sum_j \sqrt{x_j^{\text{fl}}} \leq \sqrt{kT}$. ■

The theorem makes the economics of the proof visible. The LP asks for exact aggregate counts. Static randomization produces counts with the right means, but their standard deviations are of order $\sqrt{T}$. That count mismatch is the source of the additive loss:

fluid action targets $\pm \sqrt{T}$ -scale fluctuations $\implies O(\sqrt{T})$ loss.

This is the same square-root phenomenon as in Lectures 4–5, under a different normalization: empirical means have error $O(1/\sqrt{T})$, while aggregate counts fluctuate by $O(\sqrt{T})$.

9.14 A two-value instance where the gap is real

Let T be even, $B = T/2$, and let every arrival be independently

H with probability $1/2$ and reward 2 , L with probability $1/2$ and reward 1 .

The fluid solution is $x_H^{\text{fl}} = B$ and $x_L^{\text{fl}} = 0$. Its static implementation accepts every H and rejects every L .

If $Z_H \sim \text{Binomial}(T, 1/2)$ is the realized number of high arrivals, then

$$\text{OPT} = 2 \min\{Z_H, B\} + (B - Z_H)^+,$$

whereas the high-only static policy earns $2 \min\{Z_H, B\}$. Therefore

$$\text{OPT} - R^{\text{static}} = (B - Z_H)^+, \quad \mathbb{E}[\text{OPT} - R^{\text{static}}] = \Theta(\sqrt{T}). \quad (9.19)$$

This shows that the square-root loss is real for this natural static fluid policy. It does not prove that every static policy, or every online policy, must lose $\Theta(\sqrt{T})$.

9.15 What static fluid control misses

The fluid benchmark is order T , while (9.18) loses only order $\sqrt{T}$. Its relative loss is therefore $O(1/\sqrt{T})$: the policy is first-order optimal.

But a static policy commits before learning which side of the demand fluctuation occurred. A dynamic policy sees remaining time and capacity. In the two-value example it can protect capacity for high requests early, then begin accepting low requests if too few highs have appeared.

The next question is therefore stronger:

Can a dynamic policy improve $O(\sqrt{T})$ additive loss to $O(1)$?

A point forecast of hindsight actions is not enough. We need to reason about the probability that a particular hindsight action remains desirable. That is the entry point to Lecture 11.

9.16 Takeaways

1. The offline clairvoyant is a natural upper benchmark for every online policy.
2. Relaxing integrality gives a hindsight LP whose value is concave in the demand upper bounds.
3. Concavity and Jensen give the fluid upper bound $\mathbb{E}[\text{OPT}] \leq \text{FLUID}(\mathbb{E}[\text{demand}])$.
4. The direction is not a generic property of replacing random inputs by their means; it depends on concavity of this value function.

5. The fluid LP has two outputs: primal counts predict aggregate hindsight actions, while dual variables price scarce resources and lead to bid-price controls.
6. In a network, a request must cover the combined opportunity cost of every resource it consumes.
7. A static randomized policy can implement the fluid counts in expectation by accepting type j with probability $x_j^{\text{fl}}/(Tq_j)$, subject to feasibility.
8. Undershoot and overshoot of the fluid counts have standard-deviation scale $\sqrt{T}$, yielding an explicit $O(\sqrt{T})$ additive guarantee.
9. The two-value example shows a $\Theta(\sqrt{T})$ gap for the natural high-only static fluid policy.
10. Lecture 11 asks how dynamic prediction of hindsight actions can target an $O(1)$ additive comparison.

Recommended reading

Core / closest to this lecture.

Talluri and van Ryzin, Sections 3.1.2.3 and 3.2.2–3.2.5, for bid-price controls, their limitations, and the opportunity-cost interpretation of resource prices.

Deterministic LP perspective.

Talluri and van Ryzin, Section 3.3.1. Compare the primal allocation targets with the dual resource prices; these are the two outputs emphasized in this lecture.

Course-note material.

The Jensen/concavity proofs and the explicit static-policy bound are developed in these notes. Their role is to connect the deterministic LP to the $\sqrt{T}$ stochastic scale before Lecture 11.